Q-REX

Quick-commerce data that keeps working when sessions expire, OTPs arrive and formats change.

WHY BRANDS ARE FLYING BLIND

Brands can’t reliably see their own orders, stock and sales.

Seller APIs are thin, so teams rely on browser bots that break on:

- Session expiry
- OTP prompts
- Silent format changes

WHAT WE REFUSE TO DO

Scrape with a copied token.

Internal endpoints are undocumented and change without notice, tokens expire, and bot traffic can get a seller’s account flagged. It is the fragile workflow Palm Era wants gone.

A COMPLIANT FRONT DOOR

Compliant by construction.

1

Official API

First choice wherever a platform offers one.

2

Portal export

The seller's own reports, with logged consent.

3

Report email

Scheduled reports, parsed. No session to keep alive.

4

Manual upload

The last resort, through the same checks.

- First rung that works wins
- Every record keeps its source
- OTP is always typed by a human

SEVEN STAGES, ONE PATH

Bad data never reaches the dashboard.

01	02	03	04	05	06	07
Authenticate	**Retrieve**	**Validate**	**Transform**	**Store**	**Monitor**	**Dashboard**
Encrypted sessions, expiry probe	Walks the source ladder	Drift, partial and contract checks	One SKU across platforms	Raw → staging → clean, idempotent	Freshness SLA, stock, alerts	Live updates over SSE

A failed check quarantines the pull. The clean layer is never touched.

CHAOS PANEL ON STAGE

We break it live. It recovers.

FAULT INJECTED	HANDLED OUTCOME
Column renamed	`SCHEMA_DRIFT` alert · pull quarantined · clean data untouched
HTTP 500, timeout, 429	Retries with backoff and jitter · recovers on its own
Truncated file	`PARTIAL_PULL` · quarantined · re-pulled cleanly
Session expired	Paused, seller alerted · OTP re-auth · gap backfilled
SKU runs low	`LOW_STOCK` / `STOCK_OUT` per dark store

MEASURED, NOT CLAIMED

Verified from a cold start.

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chaos scenarios handled end to end, scripted and repeatable

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duplicate rows in the clean layer, however often a sync re-runs

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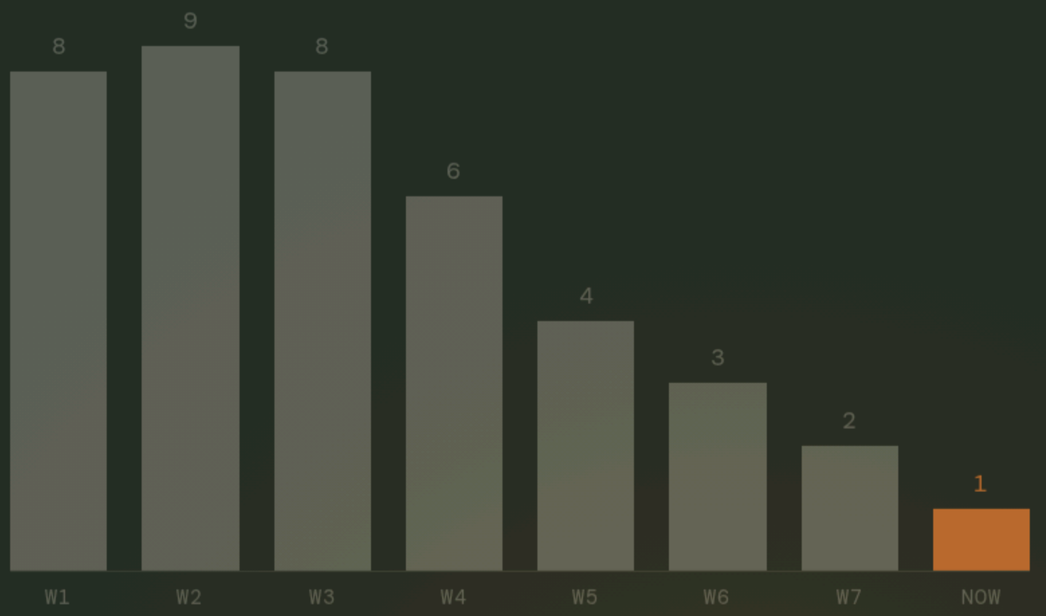
order lines recovered after a session expiry, checked against the source

HUMAN IN THE LOOP, RARELY

One tap, one code, back in sync.

- Expiry caught before the next sync
- Seller gets a one-tap re-auth link
- They type the OTP from their phone
- Q-REX resumes and backfills the gap

MANUAL INTERVENTIONS PER WEEK



Weeks 1–7 are a seeded baseline standing in for the manual workflow; “Now” is the live demo’s single re-auth.

THE BRAND'S SIDE

Every platform, one honest dashboard.

- Freshness per source and dataset
- One SKU across both platforms
- Stock alerts per dark store
- Stale data is marked, never hidden



Live screenshot · Q-REX overview running in Docker Compose

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Pull data only through permitted paths. Handle the OTP, don’t defeat it. Survive every failure, live.

Q-REX · PS-9 · PALM ERA

RUN IT YOURSELF

```
docker compose up --build
…exec api python -m scripts.seed
…exec api python -m scripts.demo
```

Dashboard :3000 · Chaos :9000 · Alerts :8025